

Gaurav Raj

Security Researcher · Software Engineer · Open Source Architect
Gurugram, Delhi, India

Mobile: +917488930330
Email: gaurav@thehackersbrain.dev
GitHub: github.com/thehackersbrain
Website: thehackersbrain.dev
LinkedIn: @thehackersbrainn
ORCID: 0009-0006-7136-5009

PROFILE

Offensive Security Researcher & Software Engineer, Published Author. Specialized in **low-level systems programming (C/ASM/Rust)**, **Linux internals**, **kernel development**, and **red team operations**. Creator and maintainer of BerserkArch Linux, focused on secure, automated, and scalable offensive security environments. Strong crossover between kernel engineering, cloud infrastructure, and full-stack development.

TECHNICAL SKILLS

Red Team & Pentest Reverse Engineering, Network/Internal/Web Pentesting, AD Attacks, C2 (Cobalt Strike/Sliver).
Tooling: NetExec, Impacket, bloodyAD, Sliver C2, BloodHound, Certipy, Ghidra, IDA Pro, BurpSuite Pro.
AI & LLM Security: LLM Fine-tuning, Prompt Engineering, Model Evaluation, Data Analysis, LLM Red Teaming.
Core Languages C, C++, Assembly, Rust, Golang, Python, Bash, Zig, JavaScript, TypeScript, Java.
Systems Engineering Linux Internals, Kernel Dev, Memory Management, Bootloaders, Device Drivers, Embedded/IoT.
Cloud & DevOps: AWS, Azure, GCP, Terraform, Vagrant, Kubernetes, Docker, Github Actions, CI/CD, Linux.

PUBLICATIONS & ACADEMIC CITATIONS

Book: Forged Trust - Offensive Operations Against ADCS. Covering deep fundamentals, ESC1-18, THEFT1-5, (D)PERSIST.
Paper: Certificate of Compromise, a full-taxonomy ADCS SoK/Whitepaper. Published on Zenodo (10.5281/zenodo.20732733).

SECURITY & ENGINEERING EXPERIENCE

BerserkArch Linux Distro Project <i>Lead Maintainer, Developer & Systems Architect</i>	Remote <i>2025 - Present</i>
OS: Designed and maintain an Arch Linux-based distribution for red team operations and secure development. CI/CD: Built fully automated ISO CI/CD pipelines using GitHub Actions and Docker (build, sign, checksum, deploy) Containerization: Published official Docker images for portable offensive security labs and rapid development. Security Hardening: Implemented custom kernel patches and AppArmor profiles for system hardening.	

Cybercraft Labs Pvt Ltd <i>Founder & Principal Engineer (Security Consultancy)</i>	Patna, Bihar <i>Sept 2023 – Nov 2024</i>
Agency Leadership: Founded and operated an independent security consultancy, managing client lifecycles from technical consultation to delivery of penetration testing and cloud security solutions. Cloud Architecture: Architected AWS/GCP hybrid infrastructures, hardened EC2, configured SES (DKIM/DMARC). Software Development: Developed full-stack web applications, desktop tools, and custom reconnaissance automation (Python/Go/Rust), optimizing client workflows and reducing manual testing time by 40%.	

KEY R&D PROJECTS

THBOS (Custom x86 Kernel) <i>Source Code</i>	<i>Systems Programming (C, Assembly & Zig)</i>
64-bit OS kernel from scratch with custom bootloader, Protected Mode, CPUID-driven topology.	
Chip8 Emulator <i>github.com/thehackersbrain/chip8</i>	<i>Systems Emulation (Rust)</i>
Functional CHIP-8 emulator with Full fetch-decode-execute cycle with memory-safe bytecode handling via borrow checker.	
HiddenWave <i>github.com/thehackersbrain/hiddenwave</i>	<i>Steganography Tool (C++ & Python)</i>
LSB audio steganography engine with WAV/MP3 support, also featured in a Netflix’s series Gyeongseong Creature S2E1.	

ACHIEVEMENTS & CERTIFICATIONS

- VulnHub:** Designed 2 vulnerable VMs (*Blogger, CorpVision*) with custom exploit chains (Webmin, Cronjob privesc).
- Advanced Security Labs:** Completed **HackTheBox Fortresses** (Jet) & **TryHackMe Networks** (Wreath, Holo), mastering pivoting, AD exploitation, and perimeter breaching.
- Global Rankings:** **Top 1% on TryHackMe** (CTF) | **HackerRank Gold Badge** (Problem Solving & Algorithms).
- Certifications:** Certified proficiency in **Computer Forensics**, **Malware Analysis**, and **Active Directory Security** (Udemy/Independent Coursework).

EDUCATION

B.Sc (Maths Hons) <i>Specialization:</i>	<i>Patliputra University, Patna</i> <i>Advanced Mathematics (ie. Calculus, Linear Algebra)</i>	2027 (ongoing) -
<i>Extensive self-directed study across red team op, adversary simulation, EDR and AV evasion, AD and windows internals.</i>		